EVERY INTROENUMERABLE SET CONTAINS A UNIFORMLY INTROREDUCIBLE SUBSET

PATRIZIO CINTIOLI

Abstract. We prove that every infinite introenumerable set contains an infinite uniformly introreducible subset, answering both parts of Question 1.7 of Greenberg, Harrison-Trainor, Patey, and Turetsky affirmatively. Moreover, a single procedure computes the original set from every infinite subset of the selected set. The proof combines the uniformization and refinement results of these authors with Ramsey-theoretic methods and a canonical reconstruction argument. We also obtain refinements controlling the ordinals and hyperjumps of the selected subsets.

1. Introduction

An infinite set A is introreducible if its characteristic function is computable from every infinite subset of A. The information carried by the set is therefore preserved under arbitrary infinite thinning, although the procedure that recovers it may depend on the subset. Uniform introreducibility requires one procedure to work for all infinite subsets. These two notions need not coincide: Lachlan's construction, presented in [5, Theorem 4.1], gives an introreducible set that is not uniformly introreducible.

A complementary direction is the study of introimmunity: for a specified reducibility, an introimmune set cannot be recovered from any subset omitting infinitely many of its elements. Recent existence results for introimmune sets with respect to several reducibilities are obtained in [1].

A standard example of a uniformly introreducible set is given by the family F_A of all finite initial segments of the characteristic function of a set A of natural numbers. Here finite binary strings are identified with natural numbers using a fixed computable bijection. It is easy to see that F_A is uniformly introreducible. Moreover, A and F_A are Turing equivalent. Thus every Turing degree contains a uniformly introreducible set; see the discussion of Dekker sets in [3, Section 1].

This recoding, however, need not produce a subset of A. The problem studied here is instead whether one can obtain uniform introreducibility by selecting an infinite subset of the original set.

The corresponding enumeration notion provides a broader setting for this problem. An infinite set is introenumerable if it is c.e. relative to each of its infinite subsets, and uniformly introenumerable if one relative c.e. operator works for all of them. Jockusch [5, p. 535] asked whether every infinite uniformly introenumerable set contains an infinite uniformly introreducible subset. Greenberg, Harrison-Trainor, Patey, and Turetsky [3, Theorem 1.4] proved that it does. They also posed the following two-part question.

Question 1.7 ([3])**.** Does every infinite introreducible set have an infinite uniformly introreducible subset? Does every infinite introenumerable set have an infinite uniformly introenumerable subset?

Subsequent work of Em gives a related sufficient condition. In [2, Proposition 4.20], an introreducible set A is shown to have a uniformly introreducible subset when it is finitely

Date: September 13, 2026.

2020 *Mathematics Subject Classification.* Primary 03D99; Secondary 05D10.

Key words and phrases. introreducibility, uniform introreducibility, introenumerability, uniform introenumerability, Ramsey theory.

decomposable, that is, when

$$A = A_0 \cup \cdots \cup A_n$$

for some infinite sets $A_0, \ldots, A_n$ such that $A \leq_T^{\mathrm{ui}} A_i$ for every $i \leq n$. Our result does not establish this finite-decomposition condition for arbitrary introreducible sets. Rather, the proof bypasses it and uses a different strategy based on uniformization, Ramsey-theoretic certification, and canonical reconstruction.

We prove that every infinite introenumerable set contains an infinite uniformly introreducible subset. This gives affirmative answers to both parts of [3, Question 1.7]. In fact, the construction provides a nonempty class of such subsets with a common reconstruction procedure, and every infinite subset of any member of the class also computes the original set A by a second common procedure. The precise statement includes a definability bound on this class, which permits separate applications of basis theorems to control the ordinal and hyperjump of a selected member.

Statement and conventions. We write $\omega = \{0, 1, 2, \ldots\}$. For a set $B \subseteq \omega$, let

$$[B]^\omega = \{D \subseteq B : D \text{ is infinite}\}.$$

We also write $[B]^{<\omega}$ for the family of all finite subsets of B, including the empty set, and $[B]^k = \{F \subseteq B : |F| = k\}$ for $k \in \omega$.

For $A \subseteq \omega$, we identify A with its characteristic function and write

$$\mathrm{Graph}(A) = \{(n, A(n)) : n \in \omega\},$$

using a computable coding of pairs. We write $A \leq_T B$ if the characteristic function of A is computable with oracle B. Fix an effective enumeration $(\Phi_e)_{e\in\omega}$ of the ordinary Turing functionals. The notation $\Phi_e^D = A$ means that Φ_e^D is total and agrees with the characteristic function of A.

For $A \subseteq \omega$ and an infinite set $B \subseteq \omega$, define

$$\begin{aligned} A \leq_T^{\mathrm{i}} B &\iff (\forall D \in [B]^\omega)\ A \leq_T D \\ &\iff (\forall D \in [B]^\omega)(\exists e \in \omega)\ \Phi_e^D = A, \end{aligned}$$

and

$$A \leq_T^{\mathrm{ui}} B \iff (\exists e \in \omega)(\forall D \in [B]^\omega)\ \Phi_e^D = A.$$

Thus $A \leq_T^{\mathrm{i}} B$ means that every infinite subset of B computes A, with the reduction index allowed to depend on that subset. In contrast, $A \leq_T^{\mathrm{ui}} B$ requires a single index that works for every infinite subset of B.

An infinite set A is called *introreducible* if $A \leq_T^{\mathrm{i}} A$, and *uniformly introreducible* (UI) if $A \leq_T^{\mathrm{ui}} A$. Explicitly,

$$A \leq_T^{\mathrm{ui}} A \iff (\exists e \in \omega)(\forall D \in [A]^\omega)(\forall n \in \omega)\ \Phi_e^D(n) \downarrow = A(n).$$

The witnessing index may depend on A, but it must be independent of the infinite subset D.

For an infinite set B, write $X \leq_{\mathrm{c.e.}}^{\mathrm{i}} B$ when X is c.e. relative to every member of $[B]^\omega$, and $X \leq_{\mathrm{c.e.}}^{\mathrm{ui}} B$ when one relative c.e. operator works for all these oracles. Write $X \leq_e^{\mathrm{ui}} B$ when one ordinary positive enumeration operator enumerates X from every member of $[B]^\omega$. An infinite set A is introenumerable if $A \leq_{\mathrm{c.e.}}^{\mathrm{i}} A$, and uniformly introenumerable if $A \leq_{\mathrm{c.e.}}^{\mathrm{ui}} A$. For infinite sets, $X \leq_{\mathrm{c.e.}}^{\mathrm{ui}} B$ and $X \leq_e^{\mathrm{ui}} B$ are equivalent by [3, Proposition 2.1]. Every introreducible set is introenumerable, and every uniformly introreducible set is uniformly introenumerable.

Every functional described as ordinary receives only the oracles explicitly indicated. All quantifiers over infinite subsets are interpreted in the ambient universe.

We use the standard lightface notation $\Sigma_1^1(A)$ and $\Delta_1^1(A)$, relativized to the oracle A; see either [7, Chapters 4–5] or [8, Section IV.2] for background.

The following theorem gives the common refinement statement discussed above. The two functionals have distinct targets: one recovers the original set A, and the other recovers the selected subset R.

Theorem 1.1. *Every infinite introenumerable set A contains an infinite uniformly introreducible subset. More precisely, there exist a nonempty $\Sigma^1_1(A)$ class $\mathcal{K} \subseteq [A]^\omega$ and two ordinary functionals Ψ, Θ such that*

$$\forall R \in \mathcal{K}\ \forall D \in [R]^\omega \qquad \Psi^D = A \quad \textit{and} \quad \Theta^D = R. \tag{1}$$

The affirmative answers to the two parts of Question 1.7 are stated separately in Corollaries 6.2 and 6.3 below. We do not assume that the final subset is hyperarithmetic in A or has the same Turing degree as A. A single decoder index works for all infinite subsets of a given R, and indeed for every R in the fixed class. We do not claim that the indices can be chosen computably from an arbitrary introenumerable set A.

For an infinite set D, let p_D be its increasing enumeration, indexed starting at zero. We identify finite sets with their increasing enumerations and write $E \preceq D$ for the initial-segment relation. We set $\max \varnothing = -1$ and $D_{>x} = D \cap (x, \infty)$. We write $\mathrm{HYP}(Z) = \Delta^1_1(Z)$ for the sets hyperarithmetic in Z, ω_1^Z for the least ordinal without a Z-computable presentation, and $\mathcal{O}^Z$ for the hyperjump of Z.

For a nonempty well-founded tree $T \subseteq \omega^{<\omega}$, we define the rank of each node $\sigma \in T$ by

$$\mathrm{rk}_T(\sigma) = \sup\{\mathrm{rk}_T(\sigma^\frown\langle x\rangle) + 1 : x \in \omega,\ \sigma^\frown\langle x\rangle \in T\}, \qquad \mathrm{rk}(T) = \mathrm{rk}_T(\varnothing).$$

Here $\sup \varnothing = 0$, so a leaf has rank zero. The rank of a rooted subtree is at most that of the containing tree.

Proof strategy and organization. The distinction between a fixed target and a selected subset is central to the argument. The uniformization theorem of GHPT [3, Theorem 1.2] turns $A \leq^{\mathrm{i}}_T B$ into $A \leq^{\mathrm{ui}}_T C$ on some infinite $C \subseteq B$. When $B = A$, this still computes A, not C. If in addition $C \in \Delta^1_1(A)$, a further refinement gives a uniformly introreducible subset [3, Lemma 2.2]. Our local criterion, Theorem 6.1, requires only $\omega_1^C \leq \omega_1^A$ in addition to $A \leq^{\mathrm{ui}}_T C$ and $C \subseteq A$; its output is a subset of A, not necessarily of C.

Section 2 combines the uniformization and refinement results of GHPT [3, Proposition 2.1, Corollary 3.13, and Proposition 5.3] with preservation of computable ordinals under hyperarithmetic reducibility, by Spector's theorem [9]; see also [7, Chapter 5].

For an infinite introenumerable A, it gives such a set C and an ordinary positive operator that enumerates $\mathrm{Graph}(A)$ from every infinite subset of C. Failures to enumerate an answer at a given input are represented by omission trees. Combining these in a single well-founded tree gives one rank bound below ω_1^A for all inputs simultaneously.

Sections 3 and 4 convert this bound into finite certificates. We construct an A-decidable front whose rank exceeds the omission bound on every infinite restriction. A binary labeling tests whether a finite approximation is correct and supplies answers on a specified finite interval. The clopen consequence of the Nash–Williams theorem [6, Theorem 1.2], together with the rank comparison, forces a nonempty homogeneous side on which these certificates are available. A decoder reads finite approximations until an answer appears; correctness is certified without requiring the decoder to recognize the front or consult A.

Section 5 addresses reconstruction of the selected subset. For a nonempty homogeneous side of a clopen coloring, relative to an oracle Z, we construct coherent color profiles on finite sets. A profile determines a canonical homogeneous set R, and Z together with any infinite $D \subseteq R$ recovers the whole profile. Repeating the canonical construction then recovers R. Arithmetical conditions on profiles and their witnesses yield a nonempty $\Sigma^1_1(Z)$ class of outputs with one common decoder.

In Section 6, we apply this relative construction with $Z = A$. The decoder for A obtained from finite certificates supplies the oracle needed to reconstruct R, giving an ordinary decoder with oracle D alone. This proves the local criterion and Theorem 1.1. The two parts of Question 1.7 follow as Corollaries 6.2 and 6.3. Separate applications of the Gandy basis theorem and the relativized hyperjump basis theorem [4, Corollary 2.7] give Corollaries 6.5 and 6.6. Section 7 concludes with open questions about the complexity of the selected subset and finite decomposability.

2. Uniformization for introenumerable sets and a single omission tree

This section supplies the initial uniform computation of A and an ordinal bound on its possible failures at finite stages. The intermediate set C is not yet required to compute itself uniformly from its infinite subsets. What matters here is that $A \leq_T^{\mathrm{ui}} C$ and that the associated omission rank is below ω_1^A. We first separate the source results for relative enumeration, positive enumeration, and computation of a characteristic function.

Uniform enumeration without ordinal increase. By [3, Corollary 3.13], for infinite sets X, B,

$$X \leq_{\mathrm{c.e.}}^{\mathrm{i}} B \implies \exists B_0 \in [B]^\omega \left(X \leq_{\mathrm{c.e.}}^{\mathrm{ui}} B_0 \ \wedge\ \omega_1^{B_0} \leq \omega_1^B\right). \tag{2}$$

Positive compilation. For infinite sets X, B, [3, Proposition 2.1] gives

$$X \leq_{\mathrm{c.e.}}^{\mathrm{ui}} B \quad \Longleftrightarrow \quad X \leq_e^{\mathrm{ui}} B. \tag{3}$$

The conversion is an effective transformation of ordinary operator indices. It neither adds an oracle nor changes the underlying set.

Improvement for an introenumerable target. By [3, Proposition 5.3], if B is infinite, then

$$\left.\begin{array}{l} X \text{ is infinite and introenumerable,} \\ X \leq_e^{\mathrm{ui}} B \end{array}\right\} \Longrightarrow \exists C \in [B]^\omega \cap \Delta_1^1(B)\ (X \leq_T^{\mathrm{ui}} C). \tag{4}$$

The introenumerability hypothesis concerns the target X, not the underlying set B. The conclusion is an ordinary uniform Turing reduction. The complexity bound is relative to B, not to X.

Lemma 2.1 (Hyperarithmetic reducibility and computable ordinals)**.** *If $X \in \Delta_1^1(Y)$, then*

$$\omega_1^X \leq \omega_1^Y, \qquad \omega_1^{Y \oplus X} = \omega_1^Y.$$

Proof. Put $Z = Y \oplus X$. Since $X \in \Delta_1^1(Y)$, closure under finite joins gives $Z \in \Delta_1^1(Y)$.

Let $\alpha < \omega_1^Z$, and choose a Z-computable presentation of a well-order of order type α. Its code is hyperarithmetic in Y, by downward closure of $\Delta_1^1(Y)$ under Turing reducibility. By Spector's theorem, relativized to Y [9] (see also [7, Chapter 5]), α has a Y-computable presentation. Hence $\alpha < \omega_1^Y$. It follows that $\omega_1^Z \leq \omega_1^Y$.

The reverse inequality follows from $Y \leq_T Z$, so $\omega_1^{Y \oplus X} = \omega_1^Y$. Finally, $X \leq_T Z$ gives $\omega_1^X \leq \omega_1^Z = \omega_1^Y$. □

Why the improvement supplies an ordinary decoder. We recall the mechanism of [3, Proposition 5.3]; its substantive input, Proposition 5.2, remains an external result. Suppose A is introenumerable and $A \leq_e^{\mathrm{ui}} B$. By [3, Proposition 5.2], choose infinite $X \subseteq A$ and $Y \subseteq B$ such that

$$X, Y \in \Delta_1^1(B), \qquad X \leq_T^{\mathrm{ui}} Y.$$

Fix an ordinary functional Φ witnessing the last reduction. Since A is introenumerable and $X \subseteq A$ is infinite, choose an ordinary index e with $W_e^X = A$. This is a choice of one natural-number index for the fixed set X; it does not assert uniformity for arbitrary subsets of A.

Let $(W_{e,s}^X)$ be a finite increasing X-computable approximation and put

$$g(n) = \min\{s : W_{e,s}^X \cap [0,n] = A \cap [0,n]\}.$$

This is a settling-time function and $g \leq_T X'$. Indeed, X' decides membership in $A = W_e^X$, and then finds stages at which the finitely many positive entries up to n have appeared. Hence $g \in \Delta_1^1(B)$. Choose a strictly increasing sequence (c_n) in Y with $c_n \geq g(n)$, computably in $Y \oplus g$, and put $C = \{c_n : n \in \omega\}$. The increasing enumeration also decides C, so $C \in \Delta_1^1(B)$.

For every $D \in [C]^\omega$, the ordinary procedure Φ^D computes X, and

$$p_D(n) \geq p_C(n) = c_n \geq g(n).$$

An ordinary decoder for $A(n)$ simulates the finite approximation $W_{e,p_D(n)}^X$ using Φ^D to answer its oracle queries, and returns 1 exactly when n belongs to that approximation. This terminates

and returns $A(n)$. Neither g, X', A, nor B is supplied as an additional oracle to the decoder. The sparse choice of C makes p_D supply a sufficient numerical time bound.

Lemma 2.2 (An ordinal-preserving consequence of GHPT)**.** *If A is infinite and introenumerable, there exist $C \in [A]^\omega$ and an ordinary positive enumeration operator Γ such that $A \leq_T^{\mathrm{ui}} C$ and*

$$\omega_1^C = \omega_1^A, \qquad \Gamma(D) = \mathrm{Graph}(A) \quad (D \in [C]^\omega). \tag{5}$$

Proof. Since $A \leq_{\mathrm{c.e.}}^{\mathrm{i}} A$, (2) gives an infinite $B \subseteq A$ such that

$$A \leq_{\mathrm{c.e.}}^{\mathrm{ui}} B, \qquad \omega_1^B \leq \omega_1^A.$$

By (3), $A \leq_e^{\mathrm{ui}} B$. Since the target A is introenumerable, (4) now gives

$$C \in [B]^\omega \cap \Delta_1^1(B), \qquad A \leq_T^{\mathrm{ui}} C.$$

Lemma 2.1 yields

$$\omega_1^C \leq \omega_1^B \leq \omega_1^A.$$

Taking $D = C$ in $A \leq_T^{\mathrm{ui}} C$, we obtain $A \leq_T C$ and hence $\omega_1^A \leq \omega_1^C$. Thus $\omega_1^C = \omega_1^A$, and $C \subseteq B \subseteq A$.

Finally, fix an ordinary Turing functional Ψ witnessing $A \leq_T^{\mathrm{ui}} C$. The ordinary relative c.e. operator which, with oracle D, enumerates (n, b) whenever $\Psi^D(n) \downarrow = b \in \{0, 1\}$ witnesses

$$\mathrm{Graph}(A) \leq_{\mathrm{c.e.}}^{\mathrm{ui}} C.$$

Since both $\mathrm{Graph}(A)$ and C are infinite, [3, Proposition 2.1] yields an ordinary positive enumeration operator Γ such that

$$\Gamma(D) = \mathrm{Graph}(A) \qquad (D \in [C]^\omega).$$

Together with $\omega_1^C = \omega_1^A$, this proves (5). □

Remark 2.3. The first use of positive compilation enumerates A, not $\mathrm{Graph}(A)$. The passage to the graph occurs only after [3, Proposition 5.3] has provided an ordinary total Turing decoder for A. Also, $C \in \Delta_1^1(B)$ does not imply $C \in \Delta_1^1(A)$: the first uniformizing set B was not asserted to be hyperarithmetic in A. Only its ordinal was controlled. The lemma alone does not show that C is uniformly introreducible: the target of $A \leq_T^{\mathrm{ui}} C$ is A, not C.

The introreducible special case. When A is introreducible, the uniformizing set can instead be obtained directly from [3, Corollary 3.13] with target $\mathrm{Graph}(A)$ and underlying set A. Indeed, every infinite subset of A computes A and hence enumerates its graph. The corollary and positive compilation give $C \in [A]^\omega$ and an ordinary Γ with $\omega_1^C \leq \omega_1^A$ and $\Gamma(D) = \mathrm{Graph}(A)$ for every $D \in [C]^\omega$. Enumerating $\Gamma(C)$ computes A, so the reverse ordinal inequality follows. This direct route does not use the introenumerable-target improvement in [3, Proposition 5.3].

We recall explicitly why the positive compilation is safe. If a computation on a finite binary string σ outputs (n, b), use the finite set $F_\sigma = \{u < |\sigma| : \sigma(u) = 1\}$ as its support. When $F_\sigma \subseteq C$, the infinite oracle

$$F_\sigma \cup \big(C \setminus [0, |\sigma|)\big)$$

is a subset of C and extends σ. Correctness on all infinite subsets forces $b = A(n)$. Thus discarding negative oracle answers introduces no incorrect outputs on supports contained in C. The computations on each infinite $D \subseteq C$ ensure that the compiled operator has enough axioms to enumerate the entire graph from that same D.

For the graph-enumeration operators used below, fix finite approximations Γ_s obtained from the first s stages of the enumeration of axioms, with $\Gamma_0(F) = \varnothing$. For finite F, the set $\Gamma_s(F)$ is finite and computable, and

$$s \leq t,\ F \subseteq E \quad \Longrightarrow \quad \Gamma_s(F) \subseteq \Gamma_t(E). \tag{6}$$

We may restrict outputs to $\omega \times \{0, 1\}$.

For every set $D \subseteq \omega$, these approximations satisfy

$$\Gamma(D) = \bigcup_{s\in\omega} \bigcup_{F\in[D]^{<\omega}} \Gamma_s(F).$$

For each input, an omission tree records finite subsets on which neither possible answer has yet appeared. We place all these trees under a common root so that their ranks have a single bound. This bound, rather than the individual component ranks, will be compared with the rank of a front in Section 4.

Lemma 2.4 (The aggregate tree)**.** *Suppose that C is infinite and that Γ is an ordinary positive enumeration operator such that $\Gamma(D) = \mathrm{Graph}(A)$ for every $D \in [C]^\omega$. For $n \in \omega$, let*

$$T_n(C) = \left\{ \tau : \begin{array}{l} \tau \text{ is strictly increasing, } \mathrm{rng}(\tau) \subseteq C, \\ (n,0),(n,1) \notin \Gamma_{|\tau|}(\mathrm{rng}(\tau)) \end{array} \right\}. \tag{7}$$

The tree

$$\mathcal{T}_\Gamma(C) = \{\varnothing\} \cup \{\langle n\rangle^\frown \tau : n \in \omega,\ \tau \in T_n(C)\} \tag{8}$$

is C-computable and well-founded. Consequently,

$$\rho := \mathrm{rk}(\mathcal{T}_\Gamma(C)) < \omega_1^C, \qquad \mathrm{rk}(T_n(C)) < \rho \quad (n \in \omega). \tag{9}$$

Proof. Since $\Gamma_0(\varnothing) = \varnothing$, the empty sequence belongs to $T_n(C)$ for every $n \in \omega$. If $\sigma \preceq \tau \in T_n(C)$, then

$$\Gamma_{|\sigma|}(\mathrm{rng}(\sigma)) \subseteq \Gamma_{|\tau|}(\mathrm{rng}(\tau))$$

by (6). Thus $\sigma \in T_n(C)$, so each $T_n(C)$ is a tree. Its membership relation is decidable in C, uniformly in n: all oracle queries concern the finitely many entries of the given sequence, and the remaining test concerns a finite computable approximation to Γ.

Fix n and suppose, towards a contradiction, that $f \in [T_n(C)]$. Then f is strictly increasing and $D = \mathrm{rng}(f)$ is an infinite subset of C. Since $\Gamma(D) = \mathrm{Graph}(A)$, there exist a finite set $F \subseteq D$ and a stage s such that

$$(n, A(n)) \in \Gamma_s(F).$$

Choose $m \geq s$ sufficiently large that $F \subseteq \mathrm{rng}(f \upharpoonright m)$. By monotonicity,

$$(n, A(n)) \in \Gamma_m(\mathrm{rng}(f \upharpoonright m)),$$

contrary to $f \upharpoonright m \in T_n(C)$. Hence every $T_n(C)$ is well-founded.

Every nonempty finite sequence has a unique representation $\langle n\rangle^\frown\tau$. The uniform decidability of the component trees therefore shows that $\mathcal{T}_\Gamma(C)$ is a C-computable tree. An infinite path through it would have a first entry n and a tail belonging to $[T_n(C)]$, which is impossible. Thus $\mathcal{T}_\Gamma(C)$ is well-founded.

The immediate successors of its root are exactly the nodes $\langle n\rangle$, for $n \in \omega$. For each n, the map $\tau \mapsto \langle n\rangle^\frown\tau$ identifies $T_n(C)$ with the subtree rooted at $\langle n\rangle$. Consequently,

$$\rho = \mathrm{rk}(\mathcal{T}_\Gamma(C)) = \sup_{n\in\omega}\big(\mathrm{rk}(T_n(C)) + 1\big).$$

In particular, $\mathrm{rk}(T_n(C)) < \rho$ for every $n \in \omega$.

Finally, let $T = \mathcal{T}_\Gamma(C)$. The Kleene–Brouwer ordering of T, with proper extensions preceding their prefixes, is a C-computable well-order. Write

$$\alpha = \mathrm{ot}(T, <_{\mathrm{KB}}) < \omega_1^C.$$

For $\sigma \in T$, let

$$h(\sigma) = \mathrm{ot}\big(\{\tau \in T : \tau <_{\mathrm{KB}} \sigma\}, <_{\mathrm{KB}}\big).$$

Every immediate successor of σ in the tree precedes σ in the Kleene–Brouwer ordering. Well-founded induction therefore gives $\mathrm{rk}_T(\sigma) \leq h(\sigma)$ for every $\sigma \in T$. In particular,

$$\rho = \mathrm{rk}_T(\varnothing) \leq h(\varnothing) < \alpha < \omega_1^C.$$

□

Remark 2.5. We do not infer (9) merely from the individual inequalities $\mathrm{rk}(T_n(C)) < \omega_1^C$: an uncontrolled supremum could reach ω_1^C. We use a single C-decidable tree that is well-founded in the ambient universe.

3. A front whose rank survives every thinning

The later Ramsey arguments will repeatedly replace an infinite set by a smaller one. We therefore need a family of finite initial segments whose tree has high rank on every infinite restriction, not merely on its original domain. The following construction obtains this persistence from a single presentation of a well-order.

Definition 3.1. A front on an infinite set V is a family $\mathcal{F} \subseteq [V]^{<\omega} \setminus \{\varnothing\}$ with no two distinct members comparable under the initial-segment relation, such that every $D \in [V]^\omega$ has an initial segment in $\mathcal{F}$. For $H \in [V]^\omega$, let

$$\mathcal{F}|H = \{F \in \mathcal{F} : F \subseteq H\}, \qquad T(\mathcal{F}|H) = \{E : \exists F \in \mathcal{F}|H \ (E \preceq F)\}.$$

Let $\ell : \mathcal{F} \to q$ be a labeling, where $1 \leq q < \omega$, and let $c(D) = \ell(F_D)$, where F_D is the unique initial segment of D in $\mathcal{F}$. For every $H \in [V]^\omega$ and every $i < q$,

$$(\forall D \in [H]^\omega)\ c(D) = i \quad \Longleftrightarrow \quad (\forall F \in \mathcal{F}|H)\ \ell(F) = i.$$

Indeed, every $F \in \mathcal{F}|H$ is the initial segment in the front of the infinite set $F \cup H_{>\max F}$. In particular, each homogeneous side $\mathcal{H}_i$ determined by these equivalent conditions is closed under taking infinite subsets.

The initial segment in the front is unique. The tree $T(\mathcal{F}|H)$ is well-founded: a path through it would be an infinite subset of H that continues strictly beyond its own initial segment in the front. We equip $[V]^\omega$ with the topology inherited from 2^ω by identifying sets with their characteristic functions. Every finite labeling of a front induces a clopen coloring of $[V]^\omega$. We use the consequence of the Nash–Williams theorem that every clopen coloring with finitely many colors has an infinite homogeneous set, even after restriction to an infinite set H [6, Theorem 1.2]. The finite-color version follows by iterating the binary version.

Lemma 3.2 (High-rank fronts)**.** *Let V be an infinite Z-computable set, and let W be a Z-computable well-order of order type β. There is a front $\mathcal{F}$ on V, decidable in Z, such that*

$$\forall H \in [V]^\omega \qquad \mathrm{rk}(T(\mathcal{F}|H)) \geq \beta. \tag{10}$$

Proof. Adjoin a new greatest element $\top$ to W, obtaining a well-order L. Using a tagged coding, we may assume that the domain of L is a Z-decidable subset of ω and that $<_L$ is decidable in Z.

For $a \in L$ and $x \in \omega$, let

$$P(a, x) = \{b \in L : b \leq x \text{ in the natural order and } b <_L a\}.$$

This finite set can be computed in Z by examining only the natural-number codes at most x. If $P(a, x)$ is nonempty, let $d(a, x)$ be its greatest element in $<_L$.

For each initial state $a \in L$, consider the following procedure on increasing sequences from V. Start in state a. When the current state is u and the next element read is x, stop if $P(u, x)$ is empty; otherwise move to state $d(u, x)$ and continue. Let $\mathcal{F}_a$ consist of the nonempty finite increasing sequences from V on which the first stop occurs precisely at the last element.

Membership in $\mathcal{F}_a$ is decidable in Z, uniformly in a. Indeed, first check that the input is a nonempty increasing sequence from V, and then simulate the procedure for the length of that sequence. Accept exactly when the first stop occurs at its final entry. No search beyond the given finite sequence is required.

No member of $\mathcal{F}_a$ properly extends another. Moreover, an infinite increasing sequence from V on which the procedure never stops would produce an infinite strictly descending sequence of states in L. Since L is well-ordered, every such infinite sequence has an initial segment in $\mathcal{F}_a$. Thus $\mathcal{F}_a$ is a front on V.

For $a \in L$ and $H \in [V]^\omega$, write

$$T_{a,H} = T(\mathcal{F}_a|H).$$

For every $x \in H$, the infinite set $\{x\} \cup H_{>x}$ has an initial segment in $\mathcal{F}_a|H$ beginning with x. Hence $\langle x\rangle \in T_{a,H}$. If $P(a,x) = \varnothing$, then $\langle x\rangle$ belongs to $\mathcal{F}_a|H$ and is a leaf of $T_{a,H}$. If $x \in H$ and $P(a,x) \neq \varnothing$, then after reading x the procedure is in state $d(a,x)$. Consequently,

$$\{\tau : \langle x\rangle^\frown \tau \in T_{a,H}\} = T_{d(a,x),H_{>x}}.$$

Indeed, the corresponding identity holds for the front members, and taking initial segments gives the displayed identity. In particular,

$$\mathrm{rk}_{T_{a,H}}(\langle x\rangle) = \mathrm{rk}(T_{d(a,x),H_{>x}}).$$

Let

$$\eta(a) = \mathrm{ot}(\{b \in L : b <_L a\}, <_L).$$

By well-founded induction on a in L, simultaneously for every $H \in [V]^\omega$, we prove

$$\mathrm{rk}(T_{a,H}) \geq \eta(a). \tag{11}$$

If a is the least element of L, then $\mathcal{F}_a|H = [H]^1$, so $\mathrm{rk}(T_{a,H}) = 1 \geq 0 = \eta(a)$.

Now assume the claim holds for all states below a, and fix an arbitrary $H \in [V]^\omega$. For each $b <_L a$, choose $x \in H$ whose natural-number value is at least the code of b. Such an x exists because H is infinite and therefore unbounded in ω. Then $b \in P(a,x)$, so

$$b \leq_L d(a,x) <_L a.$$

Since $H_{>x}$ is infinite, the induction hypothesis and the section identity above give

$$\begin{aligned}\mathrm{rk}(T_{a,H}) &\geq \mathrm{rk}_{T_{a,H}}(\langle x\rangle) + 1\\ &= \mathrm{rk}(T_{d(a,x),H_{>x}}) + 1\\ &\geq \eta(d(a,x)) + 1\\ &\geq \eta(b) + 1.\end{aligned}$$

Taking the supremum over all $b <_L a$, we obtain

$$\mathrm{rk}(T_{a,H}) \geq \sup_{b<_L a} \big(\eta(b) + 1\big) = \eta(a).$$

This completes the induction.

Finally, $\eta(\top) = \beta$. Taking $\mathcal{F} = \mathcal{F}_\top$, we obtain a Z-decidable front on V satisfying (10) for every $H \in [V]^\omega$. □

Remark 3.3. The required conclusion is a lower bound, not an exact computation of the rank. The bound holds on every infinite H, including sets not computable in Z. We assume neither an effective system of fundamental sequences nor a list of all well-orders.

4. From the omission rank to a clopen nucleus with an ordinary decoder

We now use the rank bound to obtain finite certificates for computing A. The certificates may be recognized using A, but the decoder itself must use only its given infinite subset. The proposition separates these roles: a rank comparison proves that an appropriate homogeneous side is nonempty, while monotonicity makes every answer seen before the relevant certificate correct. This also permits later constructions to leave C while retaining a decoder for A.

Proposition 4.1 (Clopen certification)**.** *Let A be infinite. Suppose there exist $C \in [A]^\omega$ and an ordinary positive operator Γ such that*

$$\Gamma(D) = \mathrm{Graph}(A) \quad (D \in [C]^\omega), \qquad \mathrm{rk}(\mathcal{T}_\Gamma(C)) < \omega_1^A.$$

There exist a front $\mathcal{G}$ on A, decidable in A, and an A-computable binary labeling $\ell : \mathcal{G} \to \{0,1\}$ such that the homogeneous side

$$\mathcal{H}_1 = \{R \in [A]^\omega : \ell(F) = 1 \text{ for every } F \in \mathcal{G}|R\} \tag{12}$$

is nonempty. Moreover, there is an ordinary functional Ψ, which does not consult A, C, or the front, such that

$$\Psi^D = A \qquad (D \in \mathcal{H}_1). \tag{13}$$

Proof. Let $\rho = \mathrm{rk}(\mathcal{T}_\Gamma(C))$. Since $\rho < \omega_1^A$, fix an A-computable well-order W of order type β such that

$$\rho < \beta < \omega_1^A.$$

This is an existence choice of a fixed index for W. We do not claim that such an index can be obtained uniformly from A and an index for Γ.

Let $\mathcal{F}$ be the A-decidable front on A supplied by Lemma 3.2. Thus

$$\mathrm{rk}(T(\mathcal{F}|K)) \geq \beta \qquad (K \in [A]^\omega).$$

Define

$$\mathcal{G} = \{\{x\} \cup E : x \in A,\ E \in \mathcal{F},\ x < \min E\}. \tag{14}$$

Membership in $\mathcal{G}$ is decidable in A: a finite set F belongs to $\mathcal{G}$ exactly when $F \subseteq A$, $|F| \geq 2$, and $F \setminus \{\min F\} \in \mathcal{F}$. If two members of $\mathcal{G}$ are comparable under the initial-segment relation, they have the same first element; removing that element gives two comparable members of $\mathcal{F}$, which must be equal. Finally, every $D \in [A]^\omega$ has an initial segment in $\mathcal{G}$: take $x = \min D$ and the initial segment in $\mathcal{F}$ of $D_{>x}$. Hence $\mathcal{G}$ is a front on A.

For $F \in \mathcal{G}$, define

$$\ell(F) = 1 \iff \begin{cases} \Gamma_{|F|}(F) \subseteq \mathrm{Graph}(A), \\ \forall n < \min F\ \exists b < 2\ ((n,b) \in \Gamma_{|F|}(F)). \end{cases} \tag{15}$$

Both conditions concern a finite computable approximation, so ℓ is A-computable.

Nonemptiness of side 1. Apply the clopen Ramsey theorem to the labeling of $\mathcal{G}|C$. There exist an infinite $H \subseteq C$ and a color $j < 2$ such that $\ell(F) = j$ for every $F \in \mathcal{G}|H$. If $j = 1$, then $H \in \mathcal{H}_1$.

Suppose instead that $j = 0$. Fix $x \in H$ with $x > 0$. For every $E \in \mathcal{F}|H_{>x}$, the set $\{x\} \cup E$ belongs to $\mathcal{G}|H$ and has color 0. Since $\{x\} \cup E \subseteq C$, positivity gives

$$\Gamma_{|E|+1}(\{x\} \cup E) \subseteq \Gamma(C) = \mathrm{Graph}(A).$$

Thus the safety condition in (15) holds, and some input below x must be omitted. Define the finite coloring

$$\chi_x(E) = \min\{n < x : (n,0), (n,1) \notin \Gamma_{|E|+1}(\{x\} \cup E)\}$$

on $\mathcal{F}|H_{>x}$. This is well-defined and takes values in $\{0, \ldots, x-1\}$.

Apply the clopen Ramsey theorem again. We obtain $K \in [H_{>x}]^\omega$ and $n < x$ such that $\chi_x(E) = n$ for every $E \in \mathcal{F}|K$. Consequently,

$$(n,0), (n,1) \notin \Gamma_{|E|+1}(\{x\} \cup E) \qquad (E \in \mathcal{F}|K).$$

By (6),

$$\Gamma_{|E|}(E) \subseteq \Gamma_{|E|+1}(\{x\} \cup E).$$

Hence the increasing enumeration of every $E \in \mathcal{F}|K$ belongs to $T_n(C)$. Since $T_n(C)$ is closed under initial segments,

$$T(\mathcal{F}|K) \subseteq T_n(C).$$

Using Lemmas 3.2 and 2.4, we obtain the contradiction

$$\beta \leq \mathrm{rk}(T(\mathcal{F}|K)) \leq \mathrm{rk}(T_n(C)) < \rho < \beta. \tag{16}$$

Therefore $j = 1$, and $\mathcal{H}_1$ is nonempty.

Decoding without recognizing the front. Define $\Psi^D(n)$ using only D and the ordinary index of Γ. Read the elements of $D_{>n}$ in increasing order. At iteration $k \geq 1$, let F_k consist of the first k elements read and compute the finite set $\Gamma_k(F_k)$. If there is a $b < 2$ such that $(n,b) \in \Gamma_k(F_k)$, return the least such b. Otherwise continue to the next iteration.

This procedure uses neither A nor C, and does not consult $\mathcal{G}$ or ℓ. In particular, it does not test whether F_k is a leaf.

To verify correctness, fix $D \in \mathcal{H}_1$ and $n \in \omega$. Since

$$\mathcal{G}|D_{>n} \subseteq \mathcal{G}|D,$$

the infinite set $D_{>n}$ still belongs to $\mathcal{H}_1$. Let F be its unique initial segment in $\mathcal{G}$, and put $m = |F|$, so that $F = F_m$ in the notation of the algorithm.

Since $\ell(F_m) = 1$ and $n < \min F_m$, we have

$$\Gamma_m(F_m) \subseteq \mathrm{Graph}(A), \qquad (n, A(n)) \in \Gamma_m(F_m).$$

For every $k \leq m$, monotonicity gives

$$\Gamma_k(F_k) \subseteq \Gamma_m(F_m) \subseteq \mathrm{Graph}(A). \tag{17}$$

Thus the procedure halts by the m-th iteration, and every answer available up to and including that iteration is correct. It follows that $\Psi^D(n) = A(n)$. Since D and n were arbitrary, $\Psi^D = A$ for every $D \in \mathcal{H}_1$. □

Remark 4.2. We have not proved that $\Gamma(R)$ is safe for every $R \in \mathcal{H}_1$: incorrect outputs could appear after the first leaf. The proof uses precisely (17); the new program halts before proceeding beyond the relevant certificate. Simply applying Γ to all of R, without this stopping rule, would not be justified.

5. Reconstructing a solution from its color profile

The remaining task is to reconstruct a selected homogeneous set, not just to compute the original target from it. We encode the choices needed for this reconstruction in a color profile on finite sets. The proof has two complementary parts: the profile determines a canonical solution, and every infinite subset of that solution, together with the base oracle, recovers the profile. Recovery must work on all finite-set inputs, since repeating a least-candidate search also tests candidates outside the final solution. The class of canonical outputs will be analytic relative to the base oracle. The only external result used in this section is the clopen Ramsey theorem.

Theorem 5.1 (A relative UI basis for a clopen side)**.** *Let V be an infinite Z-computable set, let $\mathcal{F}$ be a front on V decidable in Z, and let $\ell : \mathcal{F} \to q$ be Z-computable, where $1 \leq q < \omega$. Fix $i < q$ and suppose that the homogeneous side*

$$\mathcal{H}_i = \{H \in [V]^\omega : \ell(F) = i \text{ for every } F \in \mathcal{F}|H\}$$

is nonempty. There exist a nonempty $\Sigma^1_1(Z)$ class $\mathcal{K} \subseteq \mathcal{H}_i$ and a single ordinary functional Ξ such that

$$\forall R \in \mathcal{K}\ \forall D \in [R]^\omega \qquad \Xi^{Z \oplus D} = R. \tag{18}$$

Proof. Fix indices for the procedures computing V, $\mathcal{F}$, and ℓ relative to Z. For $D \in [V]^\omega$, let $c(D) = \ell(F_D)$, where F_D is the unique initial segment of D in $\mathcal{F}$. For each $E \in [V]^{<\omega}$, define

$$c_E(S) = c(E \cup S) \qquad (S \in [V_{>\max E}]^\omega).$$

The map $S \mapsto E \cup S$ is continuous, so c_E is clopen.

Call $E \in [V]^{<\omega}$ terminal if some $F \in \mathcal{F}$ is an initial segment of E. Terminality is decidable in Z by checking the finitely many initial segments of E. The witnessing F is unique, and in this case c_E is constantly equal to $\ell(F)$.

Existence of a coherent profile. Fix $H \in \mathcal{H}_i$. Choose a Z-computable enumeration without repetition of all finite subsets E_s of V, with $E_0 = \varnothing$. Put $H_{-1} = H$. At stage s, apply the clopen Ramsey theorem to c_{E_s} on the infinite set $(H_{s-1})_{>\max E_s}$. Choose an infinite

$$H_s \subseteq (H_{s-1})_{>\max E_s}$$

such that c_{E_s} is constant on $[H_s]^\omega$, and denote its constant value by $\tau(E_s)$.

Choose $u_s \in H_s$ so that the sequence (u_s) is strictly increasing, and let $G = \{u_s : s \in \omega\}$. For every fixed s, all but finitely many elements of G belong to H_s. We have $\tau(\varnothing) = i$, and whenever E is terminal with initial leaf F, we have $\tau(E) = \ell(F)$.

Moreover,

$$\forall E \in [V]^{<\omega}\ \exists N \geq \max E\ \forall x \in G \quad x > N \Longrightarrow \tau(E \cup \{x\}) = \tau(E). \tag{19}$$

To prove this, fix $E = E_s$ and choose $N \geq \max E$ such that $G_{>N} \subseteq H_s$. For any $x \in G_{>N}$, let t satisfy $E_t = E \cup \{x\}$. Choose an infinite tail K of G contained in $H_s \cap H_t$ and lying above x. Then

$$\tau(E) = c_E(\{x\} \cup K) = c_{E\cup\{x\}}(K) = \tau(E \cup \{x\}).$$

Using a fixed coding of finite sets, extend τ by zero on finite sets not contained in V. Call a pair (τ, G) admissible if $G \in [V]^\omega$, the values of τ lie in q, τ is zero outside $[V]^{<\omega}$, $\tau(\varnothing) = i$, τ has the prescribed values on terminal sets, and (19) holds. The preceding argument proves that an admissible pair exists. No computability in Z is asserted for τ, G, or the sets H_s. Only the stated admissibility conditions will be used below.

Canonical reconstruction from $Z \oplus \tau$. Fix any admissible pair (τ, G). Define (r_s) recursively. At stage s, let

$$b_s = \max\big(\{s\} \cup \{r_j : j < s\}\big),$$

and choose the least $x \in V$ with $x > b_s$ such that

$$\forall E \subseteq V \cap [0, b_s] \qquad \tau(E \cup \{x\}) = \tau(E). \tag{20}$$

Put $r_s = x$. This is a finite test decidable in $Z \oplus \tau$. For each of the finitely many sets E involved, (19) supplies a threshold. Every sufficiently large element of G exceeds all these thresholds and b_s, and hence is a candidate. Thus the search for the least candidate terminates.

Let $R = \{r_s : s \in \omega\}$. The sequence is strictly increasing and $r_s > s$. The reconstruction procedure uses $Z \oplus \tau$, but does not query G.

For every finite $E \subseteq R$, we have $\tau(E) = i$. Indeed, insert the elements of E in the order of their construction. When an element r_s is inserted, the previously inserted elements lie in $V \cap [0, b_s]$, so (20) preserves the value of τ. The initial value is $\tau(\varnothing) = i$. In particular, if $E \in \mathcal{F}|R$, then E is terminal and $\ell(E) = \tau(E) = i$. Therefore $R \in \mathcal{H}_i$.

Recovering the profile from $Z \oplus D$. We describe a single procedure $\Delta^{Z\oplus D}$ on finite-set inputs. Given E, return zero if $E \not\subseteq V$. If E is terminal, return the color of its initial leaf. Otherwise put

$$m = \max E + 1$$

with $m = 0$ when $E = \varnothing$, and read

$$d_j = p_D(m + j) \qquad (j \in \omega).$$

Here $m + j$ is a position in the increasing enumeration of D, not a lower bound on the numerical value of the element read. Starting with $Q_0 = E$, let

$$Q_t = E \cup \{d_0, \ldots, d_{t-1}\}.$$

Continue until Q_t becomes terminal, and then return the color of its initial leaf.

To verify this procedure, fix $D \in [R]^\omega$. Only in the verification, write

$$d_j = r_{s_j}.$$

Because D is a subset of the range of the increasing sequence (r_s), the indices s_j are strictly increasing and satisfy

$$s_j \geq m + j.$$

Hence $\max E < s_j < r_{s_j} = d_j$. Every earlier d_h, for $h < j$, was chosen before stage s_j. It follows that

$$Q_j \subseteq V \cap [0, b_{s_j}].$$

Applying (20) at stage s_j gives $\tau(Q_{j+1}) = \tau(Q_j)$. By induction,

$$\tau(E \cup \{d_0, \ldots, d_{t-1}\}) = \tau(E) \qquad (t \in \omega). \tag{21}$$

The sets Q_t are initial segments of the same infinite set

$$E \cup \{d_j : j \in \omega\} \subseteq V.$$

Since $\mathcal{F}$ is a front on all of $[V]^\omega$, one of the Q_t is terminal. If F is its initial leaf, admissibility and (21) give

$$\ell(F) = \tau(Q_t) = \tau(E).$$

Thus the procedure terminates and returns $\tau(E)$. The same conclusion holds in the terminal and out-of-domain cases by definition. Therefore

$$\Delta^{Z\oplus D} = \tau \qquad (D \in [R]^\omega).$$

This equality holds on all finite-set inputs, not merely on finite subsets of R. This will allow us to repeat the least-candidate search, including its tests on candidates outside R.

The procedure Δ uses only $Z \oplus D$ and the fixed indices for V, $\mathcal{F}$, and ℓ. It does not query R, G, or τ, and it does not compute the indices s_j.

Now define $\Xi^{Z\oplus D}$ by repeating the canonical reconstruction, answering each query to τ by the procedure $\Delta^{Z\oplus D}$. Induction on s shows that this reproduces the same r_s: once the earlier elements have been reproduced, the bound b_s, every candidate test, and the least successful candidate are unchanged. To decide whether $n \in R$, generate the sequence until the first element at least n is reached. This search terminates since $r_s > s$. Thus

$$\Xi^{Z\oplus D} = R \qquad (D \in [R]^\omega).$$

The index of Ξ is independent of the admissible pair and of the corresponding output R.

The analytic class. Let $\mathcal{K}$ consist of all sets R for which there exist a profile τ, an infinite set G, and a sequence $r \in \omega^\omega$ such that (τ, G) is admissible, r satisfies the least-candidate recursion above, and $R = \mathrm{rng}(r)$.

Admissibility is arithmetical in Z, τ, G. In particular, the quantifier over finite sets in (19) is a number quantifier. The least-candidate recursion and the condition $R = \mathrm{rng}(r)$ are also arithmetical in the displayed parameters. Consequently, $\mathcal{K}$ is $\Sigma^1_1(Z)$.

The existence of an admissible pair proves that $\mathcal{K}$ is nonempty. For every admissible pair, the preceding verification shows that its output belongs to $\mathcal{H}_i$ and that the same functional Ξ recovers that output from $Z \oplus D$ for every infinite subset D of the output. Hence $\mathcal{K} \subseteq \mathcal{H}_i$ and (18) holds.

The front and its labeling are fixed Z-computable objects. Their validity as a front and labeling is not imposed again in the formula defining $\mathcal{K}$. Likewise, universal correctness of Ξ on all infinite subsets is a consequence of the verification, not an additional clause in that formula. □

Remark 5.2. The new set R is contained in V and preserves the homogeneous color, but we do not assert that $R \subseteq H$, where H was the initial solution. That solution is used to prove the existence of the profile. The subsequent canonical choices may lie elsewhere in V. The class $\mathcal{K}$ itself need not be closed under taking infinite subsets.

6. Composition, both parts of Q1.7, and degree control

The two preceding constructions provide complementary computations. On the homogeneous side obtained by certification, each infinite subset computes A without an additional oracle. The profile construction recovers the selected set using that subset together with A. Composing these procedures removes the base oracle. We first state the resulting local criterion, whose hypotheses do not require the ambient set to be introreducible or introenumerable; Lemma 2.2 supplies them for every infinite introenumerable set. The ordinal and hyperjump corollaries then select members of the same analytic class using standard basis theorems.

Theorem 6.1 (A local criterion with ordinal preservation)**.** *Let A be infinite. Suppose that there is $C \in [A]^\omega$ such that*

$$A \leq^{\mathrm{ui}}_T C, \qquad \omega_1^C \leq \omega_1^A.$$

Then there exist a nonempty $\Sigma^1_1(A)$ class $\mathcal{K} \subseteq [A]^\omega$ and two ordinary functionals Ψ, Θ such that

$$\forall R \in \mathcal{K}\ \forall D \in [R]^\omega \qquad \Psi^D = A \quad \textit{and} \quad \Theta^D = R.$$

In particular, every member of $\mathcal{K}$ is uniformly introreducible. The set A need not be introreducible.

Proof. The uniform reduction $A \leq^{\mathrm{ui}}_T C$ uniformly enumerates $\mathrm{Graph}(A)$ from every infinite subset of C. By the positive compilation of [3, Proposition 2.1], fix an ordinary positive enumeration operator Γ such that

$$\Gamma(D) = \mathrm{Graph}(A) \qquad (D \in [C]^\omega).$$

Lemma 2.4 gives

$$\mathrm{rk}(\mathcal{T}_\Gamma(C)) < \omega_1^C \leq \omega_1^A.$$

By Proposition 4.1, there exist an A-decidable front $\mathcal{G}$ on A, an A-computable binary labeling of $\mathcal{G}$, and a nonempty homogeneous side $\mathcal{H}_1$ such that a fixed ordinary functional Ψ satisfies

$$\Psi^D = A \qquad (D \in \mathcal{H}_1).$$

Apply Theorem 5.1 with $Z = A$, $V = A$, front $\mathcal{G}$, and color 1. We obtain a nonempty $\Sigma^1_1(A)$ class

$$\mathcal{K} \subseteq \mathcal{H}_1 \subseteq [A]^\omega$$

and a single ordinary functional Ξ such that

$$\Xi^{A \oplus D} = R \qquad (R \in \mathcal{K},\ D \in [R]^\omega).$$

If $R \in \mathcal{K}$ and $D \in [R]^\omega$, then $D \in \mathcal{H}_1$, since $\mathcal{H}_1$ is closed under taking infinite subsets. Consequently, $\Psi^D = A$ for every such R and D.

Define an ordinary functional Θ by simulating Ξ with a virtual joined oracle. Queries to the second component are answered directly using D. A query at m to the first component is answered by running $\Psi^D(m)$. If a required subcomputation diverges, the simulation does not proceed past that query. If it returns a value outside $\{0, 1\}$, the simulation diverges; otherwise its value is used as the answer to the oracle query. Thus, whenever Ψ^D is total and binary-valued, the simulation satisfies

$$\Theta^D = \Xi^{\Psi^D \oplus D}. \tag{22}$$

Fix $R \in \mathcal{K}$, $D \in [R]^\omega$, and $n \in \omega$. The computation $\Xi^{A \oplus D}(n)$ halts and therefore makes only finitely many oracle queries. Every query to its first component is replaced by a terminating computation $\Psi^D(m) = A(m)$. Hence the simulation follows the same computation, with only finite delays, and returns

$$\Theta^D(n) = \Xi^{A \oplus D}(n) = R(n).$$

The computation of $\Psi^D(m)$ does not invoke Ξ, so this simulation is not circular.

The indices of Ψ and Ξ are fixed independently of R and D, and the same is therefore true of the index of Θ. We have proved both required equalities for every $R \in \mathcal{K}$ and every $D \in [R]^\omega$. The class $\mathcal{K}$ itself is unchanged, so it remains nonempty and $\Sigma^1_1(A)$. □

Proof of Theorem 1.1. Let A be infinite and introenumerable. By Lemma 2.2, there is $C \in [A]^\omega$ such that

$$A \leq^{\mathrm{ui}}_T C, \qquad \omega_1^C = \omega_1^A.$$

Apply Theorem 6.1. □

Corollary 6.2 (The first part of Question 1.7)**.** *Every infinite introreducible set has an infinite uniformly introreducible subset.*

Proof. Every introreducible set is introenumerable, so Theorem 1.1 applies. □

Corollary 6.3 (The second part of Question 1.7)**.** *Every infinite introenumerable set A has an infinite uniformly introenumerable subset. In fact, for the class $\mathcal{K}$ in Theorem 1.1, there is a single ordinary positive enumeration operator Λ such that*

$$\Lambda(D) = R \qquad (R \in \mathcal{K},\ D \in [R]^\omega).$$

Proof. Let Θ be the ordinary functional supplied by Theorem 1.1, so that

$$\Theta^D = R \qquad (R \in \mathcal{K},\ D \in [R]^\omega).$$

Uniformly from the fixed functional Θ, enumerate an axiom $F_\sigma \mapsto n$ whenever $\sigma \in 2^{<\omega}$ and a finite computation $\Theta^\sigma(n)\downarrow = 1$ is found, where

$$F_\sigma = \{u < |\sigma| : \sigma(u) = 1\}.$$

Here a computation with finite oracle σ is required to make all its oracle queries below $|\sigma|$. This defines one ordinary positive enumeration operator Λ, independently of R.

Fix $R \in \mathcal{K}$ and $D \in [R]^\omega$. If $n \in R$, the computation $\Theta^D(n) = 1$ has a finite initial-segment witness and hence produces an axiom whose support is contained in D. Thus $R \subseteq \Lambda(D)$. Conversely, if $n \in \Lambda(D)$, take a witnessing σ with $F_\sigma \subseteq D$ and $\Theta^\sigma(n) = 1$. The set

$$E = F_\sigma \cup \big(D \setminus [0, |\sigma|)\big)$$

is infinite, is contained in $D \subseteq R$, and satisfies

$$E \cap [0, |\sigma|) = F_\sigma.$$

Thus the characteristic function of E extends σ, so $\Theta^E(n) = 1$. Since $E \in [R]^\omega$, Theorem 1.1 gives $\Theta^E = R$. Hence $R(n) = \Theta^E(n) = 1$, so $n \in R$. This proves $\Lambda(D) \subseteq R$. Together with the reverse inclusion proved above, we obtain $\Lambda(D) = R$. Since $R \in \mathcal{K}$ and $D \in [R]^\omega$ were arbitrary, the same operator Λ works for every such pair. □

Remark 6.4. The uniformly introreducible conclusion is stronger as a property of the selected witness. As existential statements for a fixed infinite A, however, having a uniformly introenumerable subset and having a uniformly introreducible subset are already equivalent by [3, Theorem 1.4]. The new step here is obtaining such a witness from a merely introenumerable A using the local criterion.

Corollary 6.5 (Ordinal control)**.** *Every infinite introenumerable set A has an infinite UI subset $R \subseteq A$ such that*

$$A \leq_T R, \qquad \omega_1^R = \omega_1^A.$$

Proof. Let $\mathcal{K} \subseteq [A]^\omega$ be the nonempty $\Sigma^1_1(A)$ class supplied by Theorem 1.1. Every member of $\mathcal{K}$ is uniformly introreducible, and $\Psi^R = A$ for every $R \in \mathcal{K}$.

By the relativized Gandy basis theorem, as stated immediately after [3, Corollary 3.13], choose $R \in \mathcal{K}$ such that

$$\omega_1^{A \oplus R} = \omega_1^A.$$

Since $\Psi^R = A$, we have $A \leq_T R$ and hence $A \oplus R \equiv_T R$. Therefore, by the invariance of ω_1^X under Turing equivalence,

$$\omega_1^R = \omega_1^{A \oplus R} = \omega_1^A.$$

As $R \in \mathcal{K}$, it is an infinite uniformly introreducible subset of A, as required. □

Corollary 6.6 (Hyperjump control)**.** *For every infinite introenumerable set A, there is an infinite uniformly introreducible set $R \subseteq A$ such that*

$$A \leq_T R \leq_T \mathcal{O}^A, \qquad \mathcal{O}^R \equiv_T \mathcal{O}^A.$$

Proof. Let $\mathcal{K} \subseteq [A]^\omega$ be the nonempty $\Sigma^1_1(A)$ class supplied by Theorem 1.1. Every member of $\mathcal{K}$ is uniformly introreducible. Taking $D = R$ in (1), we also obtain $\Psi^R = A$ for every $R \in \mathcal{K}$.

Relativizing [4, Corollary 2.7] to the base oracle A, choose $R \in \mathcal{K}$ such that

$$\mathcal{O}^{A \oplus R} \equiv_T R \oplus \mathcal{O}^A \equiv_T \mathcal{O}^A.$$

Here $\mathcal{O}^{A \oplus R}$ denotes the hyperjump of the joined oracle $A \oplus R$. In particular, $R \leq_T \mathcal{O}^A$. Moreover, $\Psi^R = A$, so $A \leq_T R$ and hence $A \oplus R \equiv_T R$. Since the hyperjump preserves Turing equivalence,

$$\mathcal{O}^R \equiv_T \mathcal{O}^{A \oplus R} \equiv_T \mathcal{O}^A.$$

Finally, $R \in \mathcal{K} \subseteq [A]^\omega$ is infinite and uniformly introreducible, as required. □

Remark 6.7. The hyperjump corollary is a separate application of a basis theorem, not a new hyperjump inversion theorem. Both parts of Question 1.7 follow from Theorem 6.1 and Lemma 2.2; the ordinal control in Corollary 6.5 uses only the Gandy basis theorem. The fact that $R \leq_T \mathcal{O}^A$ does not imply $R \in \mathrm{HYP}(A)$.

Remark 6.8. The argument does not provide a uniform procedure for selecting the decoder indices or a $\Sigma^1_1(A)$ code for the class $\mathcal{K}$ from A. The final subset need not be contained in either of the intermediate sets B, C from Lemma 2.2: the local criterion reconstructs a subset of the original set A.

7. Conclusions and further questions

We have shown that every infinite introenumerable set contains an infinite uniformly introreducible subset, giving affirmative answers to both parts of [3, Question 1.7]. The proof separates uniform computation of the original set from reconstruction of the selected subset, and then composes the two procedures. The definability of the resulting class of witnesses also permits separate refinements controlling computable ordinals and hyperjumps.

A natural direction is to sharpen the complexity of the selected subset. Can the set R in Theorem 1.1 always be chosen with $R \in \Delta^1_1(A)$, while retaining $A \leq^{\mathrm{ui}}_T R$ and $R \leq^{\mathrm{ui}}_T R$? For introreducible A, one can further ask whether $R \equiv_T A$ can always be achieved, as in [2, Open Questions 4.17(2)]. The ordinal and hyperjump bounds established here do not by themselves yield either strengthening.

The argument also leaves unresolved whether every introreducible set is finitely decomposable in the sense discussed in the introduction; see [2, Open Questions 4.17(4)]. Our construction produces a suitable infinite subset without producing a finite cover of the original set by infinite sets A_i satisfying $A \leq^{\mathrm{ui}}_T A_i$. Thus the existence of a uniformly introreducible subset is settled here independently of the finite-decomposition problem.

Acknowledgments

The main result was discovered by ChatGPT 6 Pro (OpenAI). The author takes full responsibility for the mathematical content.

References

[1] P. Cintioli, *A wtt-introimmune set in Π^0_1 and introimmunity for several reducibilities*, arXiv:2603.14264v2, 2026.
[2] T. Em, *On Sets That Encode Themselves*, arXiv:2602.11134v1, 2026.
[3] N. Greenberg, M. Harrison-Trainor, L. Patey, and D. Turetsky, *Computing sets from all infinite subsets*, Trans. Amer. Math. Soc. **374** (2021), no. 11, 8131–8160. Preprint: arXiv:2011.03386v1 (numbering of the cited results follows this version).
[4] H. R. Jananthan and S. G. Simpson, *Turing degrees of hyperjumps*, arXiv:2101.08818v2, 2024.
[5] C. G. Jockusch, Jr., *Uniformly introreducible sets*, J. Symbolic Logic **33** (1968), no. 4, 521–536.
[6] A. Marcone and M. Valenti, *The open and clopen Ramsey theorems in the Weihrauch lattice*, J. Symbolic Logic **86** (2021), no. 1, 316–351.
[7] A. Montalbán, *Computable Structure Theory: Beyond the Arithmetic*, Perspectives in Logic, Cambridge University Press, 2026.
[8] P. Odifreddi, *Classical Recursion Theory. The Theory of Functions and Sets of Natural Numbers*, Studies in Logic and the Foundations of Mathematics, vol. 125, North-Holland, Amsterdam, 1989.
[9] C. Spector, *Recursive well-orderings*, J. Symbolic Logic **20** (1955), no. 2, 151–163.

Mathematics Division, School of Science and Technology, University of Camerino, Italy
Email address: patrizio.cintioli@unicam.it